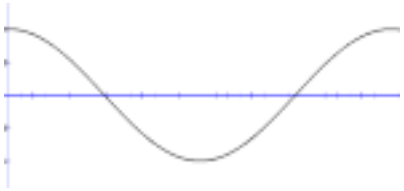
Question	Answer	Marks	Partial Marks
15	116°	**B1**	
	alternate segment theorem	**B1**	
	angles in opposite segments are supplementary or cyclic quadrilateral or angles at a point on a straight line	**B1**	
16	$8y^2 - 42y + 10[=0]$ or $8x^2 + 14x - 400[=0]$	**M3**	**M1** for $(7-3y)^2 - y^2 = 39$ oe or $x^2 - \left(\frac{7-x}{3}\right)^2 = 39$ oe **M1** for $49 - 21y - 21y + 9y^2$ or better or $49 - 7x - 7x + x^2$ or better or for correct expansion of their quadratic binomial
	$(8y-2)(y-5)[=0]$ oe $(8x-50)(x+8)[=0]$ oe	**M1**	**M1** for correct method to solve *their* quadratic equation e.g. factors, quadratic formula, completing the square
	$x = 6.25$ oe $y = 0.25$ oe $x = -8$ $y = 5$	**B2**	**B1** for $x = 6.25$, $x = -8$ or for $y = 0.25$, $y = 5$ or for a correct pair of x and y values
17	$[y =] -\frac{1}{6}x + \frac{11}{2}$ oe	**4**	**M1** for [gradient of AB =] $\frac{5 - -7}{3-1}$ oe **M1** for [gradient of perpendicular =] $-\frac{1}{\textit{their}\text{ grad } AB}$ **M1** for substituting (3, 5) in *their* linear equation
18	22.5 nfww	**3**	**M2** for $\frac{146.2 + 0.05}{7 - 0.5}$ or **M1** for 146.2 + 0.05 or 7 – 0.5 or better seen
19(a)	Correct sketch 	**2**	Needs all three features for 2 marks: • Correct curve shape • Maximum at (0, 1) and at (360, 1) and minimum at (180, – 1) • Passing through (90, 0) and (270, 0) only **B1** for two correct features

Question	Answer	Marks	Partial Marks
19(b)	75.5 or 75.52… and 284.4 to 284.5	**3**	**B2** for one correct or **M1** for $\cos x = \frac{1}{4}$ oe If 0 scored, **SC1** for two answers with a sum of 360
20	[a =] 36 [b =] – 6	**2**	**B1** for each or **SC1** for correct answers reversed
21	X, Y and Z are collinear oe	**1**	Allow in a straight line
	X is the midpoint of ZY oe	**1**	Allow e.g. $ZY = 2XY$, $ZX = XY$ oe